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RESEARCH ARTICLE

Periodicity of sex worker touring: an image-assisted analysis

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Abstract
Previous research has emphasized male tourists paying local women for sex while abroad. Recent studies reveal a shift: sex workers themselves travel on short tourist visas to provide services, often returning multiple times yearly to the same locations. Our research examined 19,608 posts and 49,008 pictures from a popular forum in Macao, a gambling hub in Asia. Through combined text mining and facial recognition analysis of online solicitations, we uncovered distinct patterns in sex worker visits. Sex workers showed significantly higher likelihood of returning within 7-day periods, aligning with Macao's visa policies. We built social networks using recognized sex workers, phone numbers, and posting accounts. This allowed us to develop metrics that assess organized crime involvement in online solicitation. Results revealed escalating third-party involvement throughout the study period. Professional brokers increasingly operated posting accounts, while shared phone numbers became more prevalent. Our analysis used a 30-day sliding window to examine recurrence factors. This approach confirmed both the 7-day periodicity and strong positive correlations between organized crime metrics and sex worker return rates.

Keywords Sex worker touring · Text mining · Image analysis · Recurrence pattern

1 Introduction

Sex workers routinely travel across locations to find work, creating different travel habits in each country [10, 19, 20, 31]. This mobility brings numerous risks. Beyond health dangers, violence, and drug addiction, workers in unfamiliar settings face higher exploitation and abuse risks [18]. The internet has transformed sex markets

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fundamentally [48]. Sex workers now rely heavily on websites to advertise services [50] and evaluate potential clients [42, 49]. Yet this digital shift has fractured worker communities into isolated individuals. Their isolation exposes them to greater control by brokers and criminal networks [17].

Macao stands as a prime location for studying these dynamics. This Special Administrative Region (SAR) of China houses 688,300 residents within just 32.9 km^2 [37]. Its laws do not criminalize prostitution itself.[1] Most sex workers arrive from Mainland China, Vietnam, and Thailand [6, 21]. Local sex trade has migrated from streets and brothels to digital spaces, mirroring global trends. Websites and mobile apps now dominate commercial sex advertising, creating new business models that attract both workers and clients [12].

Research on Macao's sex workers faces significant obstacles. Chief among these are worker mobility and shortcomings in traditional research approaches [61]. Macao's open entry policy, allowing anyone with a valid visa or departure ticket to enter, creates difficulties tracking workers' conditions, health, and safety. Past studies relied heavily on interviews with volunteer sex workers or non-random survey samples. These methods produced small sample sizes and data quality depended entirely on participant honesty [3]. Without clear data on migration trends, officials struggle to assess current regulations or implement effective HIV/STD control. The shift to internet-based sex work offers researchers new opportunities. Web-scraping and image analysis techniques now allow examination of digital footprints left by workers who typically remain hidden from conventional research methods.

Our study breaks new ground by applying these computational techniques to study sex worker movement in Macao and third-party criminal involvement in online solicitation. We combined text mining with facial recognition to track sex workers through their photos on a popular solicitation forum. By analyzing the resulting social networks, we developed metrics that measure organized crime involvement and assessed how these network characteristics relate to worker movement across time.

This research represents the first large-scale analysis of Macao's sex worker mobility using online texts and images. It fills critical knowledge gaps about hidden populations while advancing theoretical work on criminal networks and human movement. Previous research lacked detailed tools to examine power positions, coordination, and digital interactions within these networks. Our approach operationalizes network roles by mapping actor positions, creating a method to understand sex tourism networks. This helps explain how network features like centralization, brokerage, and group cohesion shape worker mobility across borders.

[1] See Lei n.º 6/97/M, https://bo.io.gov.mo/bo/i/97/30/lei06.asp for details.

2 Background

2.1 Sex worker touring

Sex workers trade sexual services or performances for money or material benefits [4]. What separates them from trafficking victims is choice—they enter transactions voluntarily, offering either direct physical contact or indirect stimulation [57]. Many come as immigrants from various regions or countries, using both legal and informal channels to enter the sex industry [59].

These workers often function as "economic migrants" [22], moving from poorer areas to seek improved prospects domestically or internationally [40]. Unlike permanent migrants, they frequently travel between locations for work [10]. Their movement follows distinct routes shaped by local laws, economic conditions, and social factors [19].

Working in unfamiliar settings creates both economic opportunities and dangers. Sex workers build cooperative networks to handle violence and mental health challenges [1, 2], seek mutual support, and resolve problems. Yet these same conditions can expose them to exploitation and abuse [18]. Criminal organizations frequently manipulate sex workers, leading authorities to classify them as victims [54].

The boundary between voluntary work and trafficking remains contested. Despite ongoing debates about defining trafficking victims and appropriate protections, evidence shows third-party transport many migrant sex workers to their destinations. These workers typically operate in brothels or similar establishments with organized crime connections [13].

Sex tourism research traditionally focused on male travelers who pay local women for sex during foreign trips [35]. This narrow view has now expanded. Recent studies show that sex workers themselves act as tourists, traveling internationally on short-term visas to provide services, often returning to profitable locations several times yearly [62]. Pratt and Suntikul [43] documented the rise of "nomadic sex workers" who strategically visit places with visa-free policies like Hong Kong. Unlike trafficking victims, these workers choose their destinations and control their movements. Many possess advanced education and maintain professional "white-collar" careers in their home countries. Their existence highlights how global economic disparities drive the worldwide spread of sex tourism [21].

2.2 Sex work in Macao

Macao sits on the western edge of the Pearl River Delta, just 65 km from Hong Kong. The region legalized gambling in 1962 while under Portuguese rule. After returning to Chinese sovereignty in 1999, Macao's casinos boomed, now generating 80% of all tax revenue [32]. Prostitution itself remains legal unless it involves brothel operations, procurement, or minors.[2] Yet all supporting activities, particularly

[2] See Lei n.º 8/2017, https://bo.io.gov.mo/bo/i/2017/26/lei08.asp for details.

client solicitation, remain illegal. This contradiction makes legal sex work practically impossible [13]. The sex trade has nevertheless flourished alongside gambling, drawing workers globally and contributing substantially to the local economy [16].

Sex workers in Macao primarily come from Mainland China, Vietnam, Thailand, and Russia [6]. Visa rules create a stratified system with unequal protections. Most Chinese workers enter on tourist visas allowing an one-time seven-day stay. Though these visas cannot be renewed within 90 days, workers can exit and re-enter with a flight ticket and visa to another country to secure another week [36]. Foreign workers from Vietnam, Thailand, and Russia often receive 3-month entertainer visas. Some acquire business visas through third-party, permitting year-long access with repeated entries [6].

These visa disparities create tiered working conditions. While sex services remain technically forbidden, entertainer and business visa holders can legally work in nightclubs and saunas. Chinese workers, barred from these venues, often resort to street solicitation, facing greater health and safety risks [6]. Working arrangements also vary widely. Some independent workers, particularly repeat visitors, manage their own travel and client acquisition [21]. Others rely on criminal networks that handle recruitment, documentation, housing, client solicitation, and supplies [13, 33].

A central debate persists over whether to classify sex workers as trafficking victims [59]. This argument stems from two opposing feminist perspectives. The "abolitionist" view [38] considers prostitution an inherent human rights violation that cannot receive genuine consent. Under this framework, all migrant sex workers qualify as trafficking victims, and anyone helping transport or arrange their work becomes a trafficker [11]. Conversely, the "sex work" approach [5] treats prostitution as legitimate labor that individuals may freely choose. This perspective draws sharp lines between trafficking victims (who face coercion and exploitation) and voluntary sex workers [7].

These theoretical disagreements produce concrete law enforcement consequences. During brothel raids, police typically arrest sex workers for document forgery, smuggling, or visa violations [13]. Officials rarely assess whether these individuals might qualify as trafficking victims deserving protection [27, 28]. Visa policies compound these problems. Chinese workers face heightened health risks, violence from clients, and arrest compared to foreign workers who operate in more protected settings like nightclubs and saunas [6].

2.3 Prostitution soliciting transitioning to online

Throughout history, sex workers found clients on streets and in public spaces. These encounters typically occurred in specific areas known as red-light districts. Street solicitation exposed workers to numerous hazards—physical violence, sexual attacks, harassment, harsh weather, and legal troubles [1, 2, 47]. According to Sanders [47], fear of violence causes the most anxiety among sex workers. Research by Okal et al. [41] shows that Kenyan sex workers must constantly avoid threats from both clients and police. Kong [26] noted that Hong Kong sex workers, operating

under laws similar to Macao's, face primary challenges of social stigma, constant surveillance, and workplace dangers.

Despite these risks, street workers developed coping strategies and maintained some control over their work [41]. However, internet access has sparked a major shift in solicitation practices. Growing numbers now seek clients online instead of on streets [50]. This change represents more than just relocation of existing activity.

Cunningham and Kendall [8] found that online solicitation has expanded the entire prostitution market rather than simply moving street activity to websites. Taylor and Canary [55] discovered that digital platforms give sex workers greater selection power. Workers can screen potential clients more thoroughly when operating online, exercising more choice and discretion. Recent studies confirm that online solicitation reduces workplace violence risks [25]. Yet this digital shift brings its own problems—workers often become more isolated and face different forms of exploitation [17].

Nomadic sex workers increasingly visit regions with visa-free policies [21, 43]. These self-employed providers depend heavily on digital platforms to find clients [9, 39]. Criminal networks have also adapted their operations. While still trafficking victims into brothels, these syndicates now offer expanded services: housing, meals, supplies, online advertising, and cross-border transport coordination [33, 34]. This shift has pushed solicitation firmly online. A recent police raid in Macao uncovered a syndicate that had operated for 10 years with specialized roles—recruiting through messaging apps, running paid forum advertisements, and delivering sex work supplies [14].

Researching sex worker movement presents numerous methodological obstacles [53]. The secretive nature of these activities hampers data collection, as participants actively avoid detection [55]. The sensitive, often illegal subject matter demands careful ethical protocols [24]. Criminal networks constantly evolve their tactics, hierarchies, and operations, making current research methods quickly obsolete [44, 60]. Cross-border operations add further complications, spanning different legal systems [58], cultural contexts, and enforcement approaches [23].

Our study tackles these challenges through innovative data collection. We gathered solicitation advertisements from a popular Macao online forum to create temporal social networks. This approach yielded new insights into sex worker movement across time. By mapping connections between posting accounts, contact phone numbers, and worker appearances, we developed metrics that quantify criminal organization involvement. These metrics allowed us to examine how criminal network features correlate with sex worker travel habits.

3 Data and methods

3.1 Data

We collected 19,608 posts containing 49,008 images related to sexual services from a prominent Macao online forum between June 2013 and October 2019. This forum launched in 2013 with a dedicated section where users shared experiences with sex

workers, posted photos, and provided contact information. The platform quickly evolved into a digital marketplace connecting workers with clients through user-created advertisements describing services, prices, and locations. Posts typically contained worker photos alongside service descriptions, pricing, and contact numbers. Users employed specialized jargon for sensitive details like prices and phone numbers to maintain privacy and avoid scrutiny. Before police shut down the operation in 2022, the forum had published over 52,000 solicitation advertisements, establishing itself as Macao's primary digital hub where sex workers, clients, and intermediaries conducted business [14]

3.2 Identifying sex workers' visits

Our study's central challenge involved tracking sex workers through online posts. Post creators rarely self-identify as workers, forcing us to use indirect methods for recognition. We relied on four data points: posting accounts, worker nicknames, contact phone numbers, and portrait photos. This approach faced additional complications when third-party (including criminal organizations) managed multiple workers, as they often used the same accounts and phone numbers across different workers. Conversely, individual workers sometimes appeared under various nicknames, phone numbers, and account names.

We solved this identification problem through facial analysis technology. We employed CompreFace [15], an open-source application built on the FaceNet model [52], to process portrait images. We ran this software locally to prevent data breaches. The system extracted facial features from each portrait by creating a 512-dimensional vector "embedding" for every detected face. These embeddings function as mathematical fingerprints with the Euclidean distance between any two vectors indicates facial similarity.

Our validation process began with 300 randomly selected posts. We manually grouped faces within these posts, assuming all faces in a single post represented the same person. After processing all images, we extracted 32,838 valid faces from the original 49,008 pictures. We then applied DBSCAN clustering to group similar faces, testing various thresholds against our validation set. Our final threshold achieved a 0.8089 recall rate, sufficient for our research needs. After removing posts without phone numbers, our dataset contained 10,476 valid posts representing 7133 distinct sex workers, 7059 phone numbers, 2023 nicknames, and 15,157 user accounts (837 posters and 14,989 repliers).

3.3 Measuring the involvement of organized crime or third-party

We built three types of social networks to uncover criminal organization involvement in online solicitation. Each network captured different relationships between key entities:

1. *Post-reply networks*: Nodes represent posting accounts, with edges connecting accounts when one replies to another.

2. *Phone-worker networks*: Nodes represent sex workers, with edges linking workers who share the same phone number.
3. *Account-worker networks*: Nodes represent sex workers, with edges connecting workers who appear in posts from the same account.

These networks allowed us to create metrics that assess criminal organization involvement. In the post-reply network, we calculated an *Agent score* that estimates whether an account belongs to a professional broker rather than a regular user. This score divides the number of posts (N_p) by the total message count within the last 30 days (posts plus replies, $N_p + N_r$):

$$S_{agent} = \frac{N_p}{N_p + N_r}$$

We conceptualize the *Agent score* as a reliable broker detector. Accounts that predominantly post without replying score near 1, strongly suggesting professional brokers. Accounts that mostly reply score near 0, indicating regular users.

We then created a second metric called the *Puppet score* to detect fake engagement. This score reveals whether a post attracts replies from suspected "sock puppet" accounts, which are fake accounts controlled by the poster to create artificial popularity. When a post receives replies, we can map connections between the original poster and responders. The score works as follows:

Let R represent all accounts that replied to a post made by account c. D_R denotes all account pairs that R created in the past 30 days. $D_{c,R}$ represents account pairs that only interact with account c. Since $D_{c,R}$ is a subset of D_R, this proportion ranges from 0 to 1:

$$S_{puppet} = \frac{|D_{c,R}|}{|D_R|}$$

When no accounts reply, $D_{c,R}$ equals 0. When all replying accounts interact exclusively with account c within the 30-day window, both numerator and denominator become equal, pushing the score toward 1. When most repliers interact broadly with many accounts beyond c, the numerator approaches 0, lowering the *Puppet score*. This metric effectively captures the use of fake accounts as a marketing strategy.

Both the *Agent score* and *Puppet score* help detect when sex workers' online presence is managed by third-party. Figure 1A visually demonstrates these concepts. Example A1 shows a high *Puppet score* scenario where all replies come from accounts that only engage with the current poster, suggesting orchestrated activity. In contrast, Example A2 shows lower scores indicating more authentic engagement from genuine users. Together, these metrics reveal when sex worker appearances are deliberately promoted through coordinated posting strategies rather than organic client interest.

Our third metric, the *Non-broker score*, identifies self-managed workers. Sex workers with personal accounts rarely promote other workers, as doing so

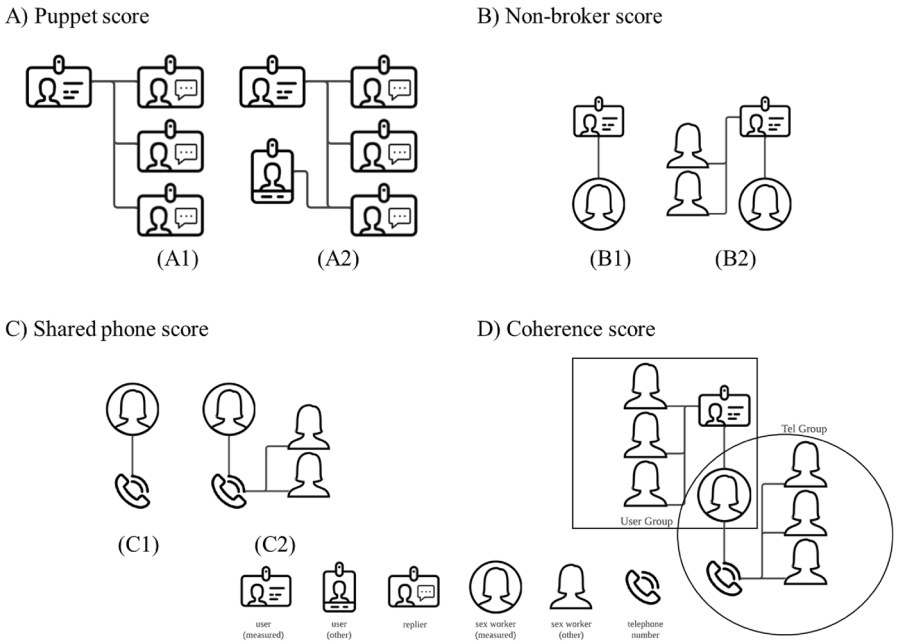

Fig. 1 Prototypical illustration of proposed scores

undermines their business. When an account posts about only one worker consistently, it likely belongs to that worker or a dedicated client.

In the account-worker network, we calculate this score using two sets: W_c represents all sex workers who appeared in posts by account c over the past 30 days, while w_p contains workers in the current post. The formula is:

$$S_{non-broker} = \frac{|W_c \cap w_p|}{|W_c| \times |w_p|}$$

This score reaches 1 only when W_c equals w_p and both contain exactly one worker. As an account posts about more workers, the denominator grows larger and $S_{non-broker}$ drops toward 0. Figure 1B illustrates this concept: example B1 shows high *Non-broker scores* where accounts focus on single workers, while B2 shows low scores where accounts promote multiple workers, suggesting third-party management.

Our fourth metric, the *Shared phone score*, detects when criminal groups provide communication tools to multiple workers. Police reports confirm that syndicates often supply phones to workers they manage. This straightforward score counts how many sex workers use the same phone number within a 30-day period. The score ranges from $[1, +\infty)$, with higher values suggesting third-party management. Figure 1C shows two examples: C1 displays a score of 1 (one worker per phone) while C2 shows a score of 3 (three workers sharing one phone).

The fifth metric, the *Coherence score*, examines alignment between online and offline networks. This score compares two networks: the "phone number-worker"

network (showing who shares phones) and the "account-worker" network (showing who gets posted by the same accounts). The formula measures overlap between these networks:

$$S_{coherence} = \frac{|W_t \cap W_p|}{|W_t \cup W_p|}$$

where W_t represents workers in the phone-sharing network posted by account c, while W_p denotes workers in the account-posting network. This ratio ranges from [0,1]. As shown in Fig. 1 (Panel D), high overlap suggests less professional criminal management, that the same people handle both online posting and phone provisioning. When criminal operations become more sophisticated, they divide these tasks among different personnel, reducing coherence.

All five metrics use a 30-day sliding window that includes the post publication date. This approach captures the platform's rapidly changing nature, focusing on recent network behavior rather than historical patterns. The window balances recency with sufficient data for robust analysis.

3.4 Modeling recurrence of sex workers

Facial recognition allowed us to track individual sex workers' appearances across time. We analyzed these movement patterns using a discrete-time recurrent event model that measured each worker's return visits within 30-day windows. When a sex worker first appeared, we began tracking their potential returns during this 30-day period. Any appearance outside this window started a new 30-day tracking cycle.

We employed the complementary log–log (C-Log–Log) link function to analyze these recurring appearances. This model calculates the probability of a worker returning at specific time points, making it well-suited for analyzing varied hazard patterns in discrete-time survival data [51]. The approach closely approximates the Cox proportional hazards model, with directly comparable coefficients [56].[3]

Since each sex worker could appear multiple times, their visits were not independent events. Unobserved personal factors likely affected their return probability. We addressed this through a Generalized Estimating Equation (GEE) approach that accounted for correlations between appearances by the same worker.[4]

The model takes this form:

$$\log(-\log(1 - \lambda(t_j|\mathbf{x}_i))) = \alpha_j + \mathbf{x}_i\boldsymbol{\beta}$$

where $\lambda(t_j)$ represents the hazard at time t_j, $\mathbf{x}_i$ is the vector of covariates for individual i, and α_j represents the baseline logit hazard for each possible event time t_j, independent of the effects of covariates $\boldsymbol{\beta}$.

[3] The results obtained from different choices of link functions, such as Logit and Probit are given in Appendix.

[4] Another reason we chose the GEE model is that the multilevel discrete-time model for recurrent events did not yield valid covariance parameter estimates in our exploratory analysis.

Our model incorporated three categories of predictors to explain sex worker movement patterns. First, we used the five network metrics described above (*Agent score*, *Puppet score*, *Non-broker score*, *Shared phone score*, and *Coherence score*) to capture criminal organization influence. Second, we added message characteristics including sentiment analysis and popularity measures (reply count, first-day reply sentiment, and self-reply frequency). Third, we examined timing variables to detect periodicity in returns after first appearance. Control variables included posting year, service location (categorized as *Hotel* or *Other*), and device type used for posting (*Cell phone* or *Other*). This comprehensive set of predictors allowed us to test how criminal network structures, online engagement patterns, and visa policies might shape sex worker movement across time.

4 Results

4.1 Descriptive analysis

Table 1 presents yearly statistics for key variables from 2014 to 2019. The forum's popularity surged dramatically during this period. Posts jumped from just 58 in 2014 to 3322 in 2018, reaching 3403 by October 2019 when data collection ended. This growth transformed how the platform functioned. The *Puppet score* that measuring fake engagement declined as the forum expanded. In 2014, this score averaged 0.51, indicating that roughly half of all replies (51%) came from accounts that only interacted with the original poster. By 2019, this indicator fell to 37%. Yet despite more posts, genuine user interaction plummeted. Average replies per post dropped from 8.0 to 0.05, while self-replies fell from 0.53 to 0.01. View counts remained stable (2746.50–2919.13), suggesting continued audience interest despite minimal interaction.

Criminal organization involvement explains these contradictory trends. The *Agent score*, which detects professional brokers, climbed from 0.83 in 2014 to 0.97 in 2019. Simultaneously, the *Non-broker score* fell from 0.88 to 0.36. These shifts reveal more broker-controlled accounts and fewer regular users posting content. Phone sharing practices also changed. The *Shared phone score* rose from 1.10 to 1.50, while the *Coherence score* dropped from 0.89 to 0.37.

These metrics tell a story of growing professionalization among sex trade networks. Early operations featured single brokers handling all aspects of sex work management, reflected in online advertising, housing, and communication tools. As the forum grew, criminal groups adopted division of labor. One team specialized in online solicitation while another managed logistics like accommodation and phone services. This split reduced overlap between posting accounts and phone networks. Rising sentiment scores in both posts and first-day replies further suggest orchestrated content creation by brokers rather than genuine user engagement.

Hotel usage patterns provide additional proof of criminal group involvement. Sex services in hotel rooms—completely absent in 2014—reached 74% of all services by 2019. Macao law explicitly bans providing sex services in hotels. This shift toward prohibited locations requires support systems that only organized groups can

Table 1 Summary statistics for the variables included by year

Year	2014	2015	2016	2017	2018	2019
	Mean(s.d.)	Mean(s.d.)	Mean(s.d.)	Mean(s.d.)	Mean(s.d.)	Mean(s.d.)
Network						
Agent	0.83 (0.28)	0.92 (0.19)	0.89 (0.21)	0.95 (0.14)	0.96 (0.12)	0.97 (0.10)
Puppet	0.51 (0.23)	0.47 (0.21)	0.49 (0.22)	0.54 (0.25)	0.47 (0.29)	0.37 (0.33)
Non-broker	0.88 (0.24)	0.81 (0.28)	0.61 (0.34)	0.46 (0.34)	0.37 (0.32)	0.36 (0.32)
Shared phone	1.10 (0.31)	1.14 (0.44)	1.19 (0.48)	1.21 (0.55)	1.33 (0.68)	1.50 (0.94)
Coherence	0.89 (0.23)	0.81 (0.27)	0.63 (0.33)	0.47 (0.34)	0.38 (0.32)	0.37 (0.31)
Message						
Msg senti	0.24 (0.79)	0.33 (0.77)	0.44 (0.72)	0.46 (0.71)	0.56 (0.66)	0.62 (0.61)
# replies	8.00 (3.96)	4.92 (4.97)	1.55 (3.51)	0.38 (1.81)	0.07 (0.71)	0.05 (0.63)
# self replies	0.53 (0.78)	0.13 (0.41)	0.14 (0.42)	0.05 (0.25)	0.02 (0.14)	0.01 (0.10)
Reply senti 1st d	− 0.19 (0.29)	− 0.18 (0.37)	− 0.16 (0.45)	0.01 (0.56)	0.31 (0.61)	0.42 (0.64)
Controls						
Device						
Cell	14 (24.1%)	188 (47.8%)	710 (58.1%)	1257 (61.3%)	1594 (48.0%)	2438 (71.1%)
Others	44 (75.9%)	205 (52.2%)	511 (41.9%)	795 (38.7%)	1728 (52.0%)	992 (28.9%)
Location						
Hotel	0 (0%)	3 (0.8%)	17 (1.4%)	331 (16.1%)	1910 (57.5%)	2539 (74.0%)
Others	58 (100%)	390 (99.2%)	1204 (98.6%)	1721 (83.9%)	1412 (42.5%)	891 (26.0%)
# of posts	58	393	1221	2052	3322	3430
# of views	2746.50 (2049.90)	3952.79 (1597.39)	3109.08 (1636.91)	2517.68 (1404.44)	2849.46 (1568.99)	2919.13 (1735.66)

provide: meal delivery, supply distribution, and lookout networks that help workers avoid police detection.

All metrics point toward the same conclusion: criminal networks steadily took control of Macao's online sex market between 2014 and 2019. This takeover appears in posting behaviors, response patterns, network structures, and the bold move toward illegal hotel-based operations. The transformation reflects a profound shift in power dynamics within this digital marketplace. What began as a diverse forum with substantial user engagement, potentially offering sex workers greater autonomy and agency in client selection and self-representation, transformed into a broker-dominated marketplace with minimal authentic interaction. This evolution from a space with characteristics of independent work toward one marked by third-party control has important theoretical implications for how we understand sex work in digital contexts, particularly regarding the debates between abolitionist perspectives that emphasize exploitation and sex work frameworks that highlight worker agency. The longitudinal patterns in our data suggest that digital platforms can shift dramatically over time, potentially supporting elements of both theoretical frameworks at different stages of their development.

4.2 Periodicity of sex worker touring

We examined sex worker return patterns using three different tracking methods: facial recognition, nickname matching, and phone number tracking. For each worker, we measured the time between first and second appearances. Figure 2 displays these return intervals with the five most prominent peaks highlighted [45].

All three tracking approaches revealed the same striking pattern: a pronounced seven-day cycle. Facial recognition data showed 7.04% of second appearances occurred exactly on day 7, with another peak of 2.99% on day 14. Both frequencies substantially exceeded typical daily return rates. Nickname tracking produced nearly identical results: 6% of returns on day 7 and 2.67% on day 14. Phone number tracking showed the strongest seven-day effect with 8.62% of returns on day 7 and 3.25% on day 14.

This seven-day cycle directly corresponds to Macao's tourist visa policy. Most mainland Chinese visitors receive seven-day entry permits. Workers holding these visas must physically leave and re-enter Macao to activate a second seven-day stay. This explains the distinct appearance spikes exactly one and 2 weeks after initial posts when workers return once their new visa periods begin.

4.3 Effects of network and message features on recurrence

We applied a discrete-time event history model to examine factors affecting sex worker return timing. Our analysis tested how network metrics (*Agent*, *Puppet*, *Non-broker*, *Shared phone*, and *Coherence* scores) and message traits (reply counts, self-replies, message sentiment, and first-day reply sentiment) influenced return probability. We controlled for location type, posting device, and year while constraining all network measures to a 30-day sliding window from each post date. For robustness

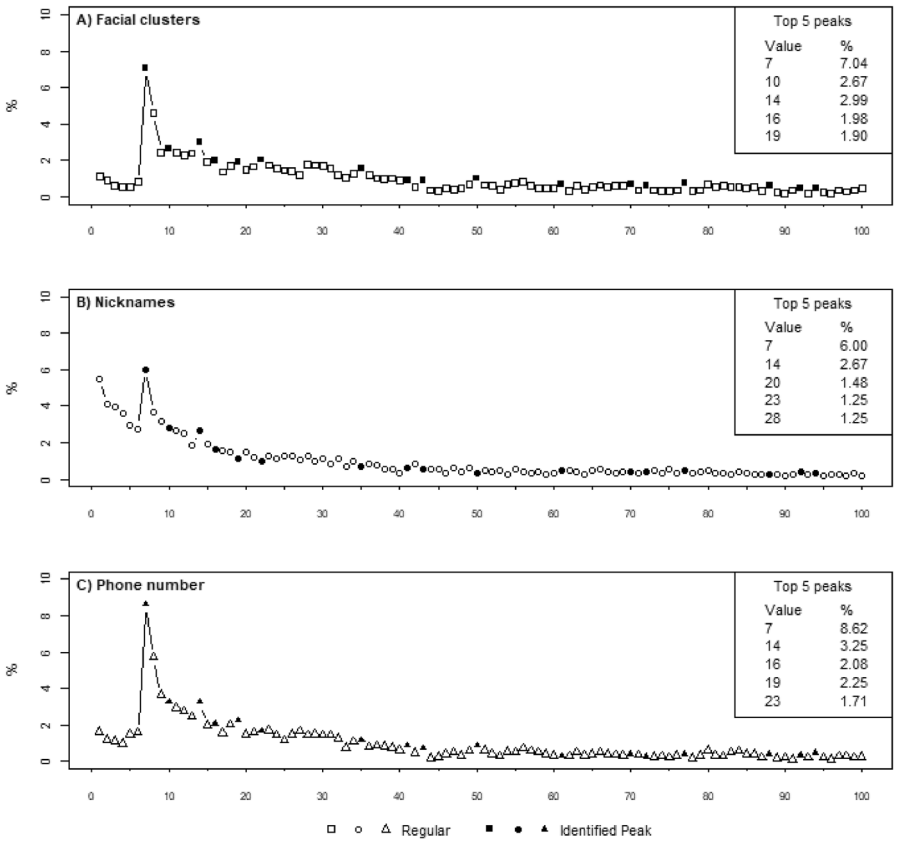

Fig. 2 The number of days apart from first occurrence based on individuals identified by facial features, nicknames, and phone numbers with marked peaks

check, we calculated 95% Bootstrapping Confidence Intervals (BCIs) from 1000 resampling iterations with 1000 observations each.[5]

Table 2 confirms the seven-day pattern seen in Fig. 1, even after controlling for all other factors. The odds of a second appearance on day-7 were 2.374 times higher (exp(1.216) − 1) than other days for workers tracked through facial recognition. This pattern persisted across subsequent returns: third visits showed 1.630 times higher odds (exp(0.967) − 1) of occurring on day-7, fourth visits 1.056 times higher (exp(0.721) − 1), and fifth + visits 1.221 times higher (exp(0.798) − 1). Day-14 also showed strong periodicity, with second appearances twice as likely (exp(0.705) = 2.02) to occur then compared to other days. While coefficients for later visits on day 14 were positive, these results didn't meet robustness thresholds.

Of the five network metrics, three showed robust effects on sex worker returns: *Non-broker score*, *Shared phone score*, and *Coherence score*. A one-unit rise in

[5] Results using different sample sizes (500, 2,000, 3,000, and 5,000) appear in the Appendix.

Table 2 Results of discrete time recurrent event model for the recurrence of sex workers identified by facial clusters within a 30-day window

Parameters	2nd vs 1st Beta[2.5th, 97.5th]BCI	3rd vs 2nd Beta[2.5th, 97.5th] BCI	4th vs 3rd Beta[2.5th, 97.5th] BCI	5th & + vs 4th Beta[2.5th, 97.5th] BCI
Intercept	− 4.402[− 4.416, − 3.495]***#	− 4.451[− 4.584, − 3.371]***#	− 4.486[− 4.828, − 3.500]***#	− 4.502[− 4.861, − 4.098]***#
Network				
Agent	− 0.375[− 0.369, 0.222]*	− 0.609[− 1.022, − 0.023]*#	− 0.196[− 0.999, 0.234]	0.212[− 0.151, 0.562]
Puppet	0.068[− 0.221, 0.146]	0.146[− 0.258, 0.072]	− 0.164[− 0.150, 0.322]+	− 0.022[− 0.275, − 0.006]#
Non-broker	− 0.974[− 2.125, − 0.814]*#	− 1.018[− 1.236, − 0.153]**#	− 1.801[− 2.172, − 0.537]***#	− 0.774[− 1.121, − 0.544]***#
Shared phone	0.491[0.332, 0.655]***#	0.618[0.428, 0.768]***#	0.708[0.528, 0.780]***#	0.527[0.458, 0.601]***#
Coherence	0.876[0.531, 1.737]+#	0.842[0.128, 1.121]**#	1.632[0.331, 1.791]***#	0.630[0.436, 0.966]***#
Message				
Msg senti	− 0.041[− 0.059, 0.072]	− 0.009[− 0.042, 0.119]	− 0.105[− 0.123, 0.022]**	0.012[− 0.014, 0.073]
# of replies	− 0.011[− 0.050, 0.031]	− 0.026[− 0.034, 0.055]+	0.000[− 0.036, 0.044]	− 0.013[− 0.035, 0.017]
# of self replies	0.229[− 0.144, 0.424]*	0.175[− 0.173, 0.328]	0.113[− 0.511, 0.201]	0.192[− 0.145, 0.208]+
Reply senti 1st d	0.007[− 0.051, 0.147]	0.148[0.006, 0.209]**#	− 0.143[− 0.166, 0.015]**	− 0.051[− 0.101, 0.000]+
Time stamp				
Day 7	1.216[0.597, 0.857]***#	0.967[0.513, 0.799]***#	0.702[0.273, 0.665]**#	0.798[0.752, 0.952]***#
14	0.705[0.090, 0.458]***#	0.166[− 0.514, 0.006]	0.607[0.256, 0.718]+#	0.212[0.156, 0.458]#
21	0.481[− 0.331, 0.183]+	− 0.204[− 0.539, 0.072]	0.341[− 0.030, 0.547]	0.255[0.101, 0.438]#
Controls				
Device				
Cell	− 0.039[− 0.213, − 0.025]#	− 0.129[− 0.147, 0.020]*	− 0.033[− 0.111, 0.108]	0.033[− 0.048, 0.091]
Others	–	–	–	–
Location				
Hotel	− 0.146[− 0.227, − 0.024]+#	− 0.002[− 0.161, 0.034]	− 0.028[− 0.069, 0.171]	− 0.037[− 0.082, 0.046]
Others	–	–	–	–

Table 2 (continued)

Parameters	2nd vs 1st Beta[2.5th, 97.5th]BCI	3rd vs 2nd Beta[2.5th, 97.5th] BCI	4th vs 3rd Beta[2.5th, 97.5th] BCI	5th & + vs 4th Beta[2.5th, 97.5th] BCI
Year				
2014	0.738[− 0.375, 1.957]*			
2015	0.006[− 0.244, 0.499]	0.453[− 0.369, 0.747]*	0.386[− 0.226, 1.233]+	− 0.179[− 0.264, 0.016]*
2016	− 0.043[− 0.084, 0.350]	0.411[0.132, 0.574]**#	− 0.163[− 0.206, 0.280]	− 0.060[− 0.181, 0.115]
2017	− 0.006[− 0.145, 0.170]	0.289[0.099, 0.427]**#	− 0.086[− 0.123, 0.204]	0.108[− 0.018, 0.157]+
2018	0.084[− 0.101, 0.212]	0.334[0.084, 0.356]***#	− 0.057[− 0.080, 0.204]	0.086[0.014, 0.170]+#
2019	–	–	–	–
QIC	9485	4853	3205	9215
QICu	9486	4865	3223	9227
N	36, 320	18, 536	12, 994	34, 299

Coefficients were estimated from the whole sample, while BCIs were obtained from 1000 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Non-broker score (indicating worker self-management) reduced second visit odds by 37.76% (exp(−0.974) = 0.3776). This reveals that independently operating workers made fewer return trips. Conversely, each additional worker using the same phone number boosted return odds by 63.39% (exp(0.491) − 1), showing that criminal-supplied phones predicted more frequent visits. *Coherence score* showed the strongest effect. A one-unit rise more than doubled return odds with a 240.2% increase (exp(0.876) = 2.402). Interestingly, as *Coherence scores* declined over the study period (Table 1), dropping from 0.89 in 2014 to 0.37 in 2019, the data suggests that growing criminal network sophistication may have reduced return frequency, particularly for third visits, where coefficients for years 2016–2018 showed robust significance (0.411, 0.289, and 0.334 respectively). This temporal pattern aligns with our observation of increasing professionalization in criminal management, where specialized roles for online solicitation versus logistics emerged, creating more complex but potentially less effective worker control mechanisms.

Message and control variables proved less influential. Neither sentiment metrics nor message popularity measures (reply counts, self-replies, first-day reply sentiment) showed robust effects on return timing. Control variables like device type and service location only reliably predicted second visits, failing bootstrapping tests for subsequent returns. This pattern reinforces our key finding: criminal network structure, not message content or superficial factors, most strongly determines sex worker movement patterns in Macao.

5 Discussion and conclusion

Our analysis of 19,608 posts and 49,008 images from a Macao forum revealed clear patterns in sex worker movement and criminal organization involvement. Using three tracking methods which focus on nicknames, facial recognition, and phone numbers, we documented a striking seven-day periodicity in worker appearances. This cycle directly matches Macao's visa policies, where mainland Chinese visitors typically receive seven-day permits. We created five metrics to assess criminal network involvement: *Agent score*, *Puppet score*, *Non-broker score*, *Shared phone score*, and *Coherence score*. These showed criminal organizations steadily gained control of online solicitation during our study period. As professional brokers increasingly managed posting accounts and shared phones among multiple workers, genuine user engagement dropped despite growing post numbers. Our statistical models confirmed that both seven-day visa cycles and criminal network metrics strongly predicted worker return timing. These findings align with police reports from a 2022 crackdown on the forum's operators [14].

Our research makes several methodological contributions to studying hidden populations. We overcame traditional sample size and data quality limitations that plague sex worker research. As stigmatized and mobile populations, sex workers often avoid academic researchers, yielding limited empirical data. Our computational approach not only tracked workers across time but captured detailed service information from solicitation posts spanning 6 years. This method offers significant advantages over traditional fieldwork and surveys. It costs less, allows repeated data

collection, and captures real-time market changes invisible to conventional research methods. The approach produces more accurate data about customer numbers, worker movement, and price changes that could inform both law enforcement and public health interventions targeting regional HIV/STD transmission.

Our research also advances criminal network detection techniques for online spaces. We created metrics that track organized crime involvement in digital sex work solicitation, a critical need as more workers and third parties move to web platforms. These tools detect criminal networks and measure their sophistication over time. Our indices revealed growing criminal organization control in Macao's online sex market, with rising professionalization and division of labor among criminal groups. These findings offer law enforcement and policymakers new ways to monitor criminal evolution in digital spaces.

We carefully addressed ethical challenges throughout our computational social science approach. While digital methods can transform knowledge creation [63], they raise ethical concerns about informed consent during data collection [29, 46]. These issues become even more complex when studying marginalized populations like sex workers. We implemented multiple safeguards: following *Fair Use* principles for web scraping [30], limiting collection to research purposes, complying with *Terms of Service* and *robots.txt* protocols, and sampling only a portion of available content. Since obtaining individual consent was impossible, we took extra privacy precautions: extracting only non-identifiable data, keeping facial features separate from other metadata, using offline recognition software to prevent data exposure, and reporting only aggregated group-level results. We further delayed analysis until 2021, 2 years after collection, and focused exclusively on group patterns rather than individual behaviors.

Our findings speak directly to the longstanding theoretical debate between abolitionist and sex work perspectives. The emergence of digital platforms has reshaped how we might understand this theoretical tension. While the abolitionist view [38] emphasizes exploitation, coercion, and victimhood, seeing all sex work as inherently harmful, our longitudinal analysis reveals a more nuanced evolution. Early in our study period (2014–2015), the forum exhibited characteristics consistent with the sex work perspective [5]—independent workers exercising agency through digital self-representation, client selection, and autonomous business management (evidenced by higher *Non-broker scores* and lower *Agent scores*). However, our network metrics document a clear trajectory toward increasing third-party control between 2016 and 2019. The rising *Agent* and *Shared phone scores* alongside declining *Non-broker* and *Coherence scores* suggest criminal networks systematically took over what may have begun as a more worker-controlled space. This evolution illustrates how digital platforms can transform from tools of worker empowerment to instruments of exploitation, challenging both theoretical perspectives. Our findings suggest neither framework fully captures the fluid reality of sex work marketplaces, which can shift dramatically over time from spaces of relative autonomy to sites of organized exploitation as criminal networks identify and capitalize on profitable digital territories. This temporal dimension adds critical context to theoretical debates that often present static conceptualizations of sex work environments.

Our study has important limitations. Without labeled facial data, we relied on assumptions to set similarity thresholds for matching faces. We presumed all faces

within a single post represented the same person and used within-post pairs as our validation set. Our optimal threshold (84) achieved a 0.8089 recall rate, but violations of our core assumption could introduce measurement errors that might distort findings. Additionally, lacking information on incorrect face pairs prevented precision calculations and F1-score evaluation of FaceNet's performance in generating the 7133 facial clusters. Future research should establish more reliable methods for validating facial recognition consistency in similar contexts.

Our findings face important generalizability constraints. Though we studied Macao's most active solicitation forum, other platforms likely exist for similar purposes. Our research represents just one scientific approach to understanding sex worker movement in this region, and readers should exercise caution with broader generalizations. Replication poses another challenge since police have shut down the forum and deleted all original content.[6] To support scientific transparency despite this obstacle, we will make our processed data available (including facial feature vectors, retrieved post content, and analysis code) to enable independent verification of our findings. Although our findings about the involvement of organized crime closely aligned with the police report, we were not able to verify whether an individual who posted the soliciting information is a broker. According to the police report, sex workers need to pay MOP300 for each post, and the criminal syndicate has managed and sold more than 52,000 advertising posts with the alleged earnings of more than MOP 15 million, which makes it the largest case in recent years. Since the forum was cracked down by police in 2022 and no more data collection is possible, we are still working on finding other source of data to validate our measures.

The computational approach also limits our insight into human factors. Our methods cannot capture crucial contextual information such as workers' socioeconomic backgrounds, working conditions, motivations, experiences, or decision-making processes. These gaps prevent us from constructing complete causal pathways for worker movement. Future research should combine our computational techniques with traditional field interviews to create a more complete picture of sex worker experiences while promoting health and safety interventions for both workers and clients.

This study breaks new ground in documenting sex worker movement patterns in Macao, a phenomenon previously hidden from researchers. Beyond its specific findings, our work demonstrates how computational social science can transform research with hidden populations. By combining text mining, image analysis, and network metrics, researchers can examine sensitive topics without the access barriers that hamper traditional methods. We hope our approach inspires more studies that apply these techniques to produce rigorous knowledge about hard-to-reach populations efficiently, accurately, and ethically.

[6] See Policia Judiciaria. (2022). Local criminal syndicate procuring prostitution busted, with 10 men and women apprehended. Retrieved June 11, 2024, from https://www.pj.gov.mo/Web//Policia/202209/20220927/13887.html?lang=en

Table 3 Results of discrete time recurrent event model for the recurrence of sex workers by logit link

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th and + vs. 4th Beta[2.5th, 97.5th] BCI
Intercept	− 4.429[− 4.478, − 3.559]***#	− 4.496[− 4.578, − 3.501]***#	− 4.523[− 4.736, − 3.585]***#	− 4.523[− 4.830, − 4.170]***#
Network				
Agent	− 0.374[− 0.352, 0.209]*	− 0.596[− 0.944, − 0.012]*#	− 0.189[− 0.924, 0.148]	0.169[− 0.149, 0.549]
Puppet	0.068[− 0.190, 0.130]	0.144[− 0.247, 0.067]	− 0.167[− 0.127, 0.283]+	− 0.017[− 0.265, − 0.030]#
Non-broker	− 0.949[− 2.103, − 0.851]+#	− 0.970[− 1.195, − 0.219]**#	− 1.767[− 2.224, − 0.583]***#	− 0.740[− 1.057, − 0.540]***#
Shared phone	0.519[0.358, 0.699]***#	0.659[0.458, 0.765]***#	0.741[0.556, 0.786]***#	0.577[0.501, 0.639]***#
Coherence	0.854[0.597, 1.709]+#	0.801[0.180, 1.070]*#	1.602[0.359, 1.835]***#	0.601[0.444, 0.931]***#
Message				
Msg senti	− 0.041[− 0.054, 0.071]	− 0.013[− 0.032, 0.110]	− 0.104[− 0.120, 0.010]**	0.007[− 0.014, 0.062]
# of replies	− 0.011[− 0.045, 0.031]	− 0.026[− 0.027, 0.050]+	0.000[− 0.031, 0.045]	− 0.014[− 0.032, 0.011]
# of self replies	0.232[− 0.118, 0.387]*	0.179[− 0.153, 0.295]	0.125[− 0.483, 0.194]	0.194[− 0.131, 0.197]+
Reply senti 1std	0.004[− 0.035, 0.131]	0.141[0.013, 0.198]**#	− 0.142[− 0.166, 0.000]**	− 0.044[− 0.093, 0.001]
Time stamp				
Day 7	1.244[0.464, 0.997]***#	0.987[0.362, 0.945]***#	0.716[0.038, 0.813]**#	0.821[0.700, 1.109]***#
14	0.714[− 0.164, 0.641]***	0.161[− 0.863, 0.240]	0.610[− 0.043, 0.892]+	0.204[− 0.035, 0.552]
21	0.485[− 0.715, 0.402]+	− 0.208[− 0.955, 0.307]	0.345[− 0.416, 0.831]	0.255[− 0.135, 0.583]
Controls				
Device				
Cell	− 0.040[− 0.207, − 0.031]#	− 0.127[− 0.140, 0.014]*	− 0.034[− 0.101, 0.091]	0.026[− 0.043, 0.080]
Others				
Location				
Hotel	− 0.147[− 0.220, − 0.033]+#	− 0.002[− 0.164, 0.026]	− 0.028[− 0.044, 0.153]	− 0.032[− 0.065, 0.039]

Table 3 (continued)

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th and + vs. 4th Beta[2.5th, 97.5th] BCI
Others				
Year				
2014	0.762[− 0.281, 1.906]*			
2015	0.007[− 0.219, 0.434]	0.442[− 0.323, 0.548]*	0.392[− 0.157, 0.996]+	− 0.168[− 0.241, 0.012]*
2016	− 0.042[− 0.049, 0.337]	0.397[0.140, 0.565]**#	− 0.163[− 0.213, 0.245]	− 0.054[− 0.143, 0.085]
2017	− 0.003[− 0.133, 0.148]	0.279[0.112, 0.409]**#	− 0.087[− 0.114, 0.202]	0.113[− 0.014, 0.155]*
2018	0.087[− 0.081, 0.181]	0.328[0.072, 0.332]***#	− 0.059[− 0.067, 0.180]	0.075[0.011, 0.157]+#
2019	–	–	–	–
QIC	9475	4847	3202	9202
QICu	9476	4860	3221	9217
N	36, 320	18, 536	12, 994	34, 299

Coefficients were estimated from the whole sample, while BCIs were obtained from 1000 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Table 4 Results of discrete time recurrent event model for the recurrence of sex workers by Probit link

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th & + vs. 4th Beta[2.5th, 97.5th] BCI
Intercept	− 2.329[− 2.391, − 1.977]***#	− 2.370[− 2.404, − 1.936]***#	− 2.381[− 2.472, − 2.008]***#	− 2.369[− 2.475, − 2.230]***#
Network				
Agent	− 0.137[− 0.141, 0.085]+	− 0.215[− 0.388, 0.010]*	− 0.062[− 0.355, 0.068]	0.045[− 0.070, 0.169]
Puppet	0.026[− 0.073, 0.053]	0.055[− 0.097, 0.027]	− 0.068[− 0.056, 0.106]*	− 0.002[− 0.103, − 0.012]#
Non-broker	− 0.322[− 0.856, − 0.320]#	− 0.335[− 0.480, − 0.064]*#	− 0.677[− 0.901, − 0.209]***#	− 0.289[− 0.433, − 0.209]***#
Shared phone	0.247[0.181, 0.348]***#	0.319[0.227, 0.357]***#	0.347[0.268, 0.372]***#	0.290[0.255, 0.315]***#
Coherence	0.293[0.236, 0.703]#	0.278[0.036, 0.450]+#	0.625[0.122, 0.788]**#	0.240[0.177, 0.386]***#
Message				
Msg senti	− 0.015[− 0.020, 0.031]	− 0.009[− 0.012, 0.041]	− 0.038[− 0.046, 0.006]**	0.005[− 0.003, 0.026]
# of replies	− 0.004[− 0.018, 0.012]	− 0.010[− 0.011, 0.021]+	0.000[− 0.011, 0.018]	− 0.006[− 0.012, 0.004]
# of self replies	0.092[− 0.043, 0.169]*	0.077[− 0.063, 0.134]+	0.073[− 0.152, 0.087]	0.085[− 0.046, 0.082]+
Reply senti 1std	− 0.003[− 0.016, 0.050]	0.047[0.005, 0.073]*#	− 0.053[− 0.072, − 0.002]**#	− 0.016[− 0.036, 0.001]
Time stamp				
Day 7	0.519[0.188, 0.442]***#	0.408[0.134, 0.418]***#	0.288[− 0.010, 0.352]*	0.358[0.294, 0.501]***#
14	0.277[− 0.085, 0.277]**	0.041[− 0.384, 0.080]	0.227[− 0.051, 0.378]	0.058[− 0.049, 0.217]
21	0.178[− 0.305, 0.161]	− 0.100[− 0.405, 0.118]	0.119[− 0.198, 0.338]	0.083[− 0.084, 0.226]
Controls				
Device				
Cell	− 0.014[− 0.081, − 0.010]#	− 0.046[− 0.058, 0.005]*	− 0.014[− 0.043, 0.032]	0.008[− 0.020, 0.030]
Others				
Location				
Hotel	− 0.054[− 0.088, − 0.010]+#	0.001[− 0.064, 0.009]	− 0.013[− 0.022, 0.059]	− 0.011[− 0.025, 0.017]

Table 4 (continued)

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th & + vs. 4th Beta[2.5th, 97.5th] BCI
Others				
Year				
2014	0.318[− 0.102, 0.894]*			
2015	0.001[− 0.089, 0.174]	0.163[− 0.126, 0.208]*	0.160[− 0.051, 0.432]+	− 0.047[− 0.073, 0.020]
2016	− 0.016[− 0.016, 0.140]	0.136[0.051, 0.221]**#	− 0.063[− 0.082, 0.099]	− 0.019[− 0.053, 0.041]
2017	0.002[− 0.054, 0.060]	0.094[0.037, 0.156]**#	− 0.034[− 0.048, 0.077]	0.052[0.003, 0.070]*#
2018	0.035[− 0.032, 0.075]	0.118[0.023, 0.125]***#	− 0.023[− 0.028, 0.071]	0.028[0.002, 0.059]#
2019	–	–	–	–
QIC	9433	4827	3189	9177
QICu	9435	4844	3211	9197
N	36, 320	18, 536	12, 994	34, 299

Coefficients were estimated from the whole sample, while BCIs were obtained from 1000 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Table 5 Results of discrete time recurrent event model with 500 bootstraps samples

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th &+vs. 4th Beta[2.5th, 97.5th] BCI
Intercept	− 4.402[− 4.788, − 3.285]***#	− 4.451[− 4.860, − 3.030]***#	− 4.486[− 5.128, − 3.167]***#	− 4.502[− 4.990, − 3.912]***#
Network				
Agent	− 0.375[− 0.547, 0.409]*	− 0.609[− 1.319, 0.176]*	− 0.196[− 1.366, 0.512]	0.212[− 0.337, 0.749]
Puppet	0.068[− 0.296, 0.242]	0.146[− 0.340, 0.157]	− 0.164[− 0.221, 0.426]+	− 0.022[− 0.341, 0.044]
Non-broker	− 0.974[− 2.465, − 0.468]*#	− 1.018[− 1.499, 0.181]**	− 1.801[− 2.564, − 0.037]***#	− 0.774[− 1.271, − 0.397]***#
Shared phone	0.491[0.310, 0.812]***#	0.618[0.394, 0.842]***#	0.708[0.493, 0.832]***#	0.527[0.436, 0.677]***#
Coherence	0.876[0.238, 2.019]+#	0.842[− 0.149, 1.365]**	1.632[− 0.182, 2.121]***	0.630[0.304, 1.115]***#
Message				
Msg senti	− 0.041[− 0.087, 0.112]	− 0.009[− 0.071, 0.152]	− 0.105[− 0.149, 0.051]**	0.012[− 0.032, 0.094]
# of replies	− 0.011[− 0.070, 0.053]	− 0.026[− 0.051, 0.094]+	0.000[− 0.054, 0.068]	− 0.013[− 0.043, 0.036]
# of self replies	0.229[− 0.258, 0.643]*	0.175[− 0.284, 0.515]	0.113[− 0.836, 0.309]	0.192[− 0.236, 0.333]+
Reply senti 1std	0.007[− 0.089, 0.192]	0.148[− 0.041, 0.247]**	− 0.143[− 0.204, 0.058]**	− 0.051[− 0.129, 0.024]+
Time stamp				
Day 7	1.216[0.248, 1.064]***#	0.967[0.098, 1.038]***#	0.702[− 0.259, 0.938]**	0.798[0.535, 1.147]***#
14	0.705[− 0.503, 0.780]***	0.166[− 1.417, 0.450]	0.607[− 0.483, 1.058]+	0.212[− 0.232, 0.688]
21	0.481[− 1.205, 0.593]+	− 0.204[− 1.652, 0.544]	0.341[− 1.026, 1.048]	0.255[− 0.431, 0.728]
Controls				
Device				
Cell	− 0.039[− 0.247, 0.016]	− 0.129[− 0.197, 0.065]*	− 0.033[− 0.157, 0.159]	0.033[− 0.080, 0.117]
Others				
Location				
Hotel	− 0.146[− 0.277, 0.024]+	− 0.002[− 0.213, 0.074]	− 0.028[− 0.117, 0.225]	− 0.037[− 0.101, 0.072]

Table 5 (continued)

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th &+vs. 4th Beta[2.5th, 97.5th] BCI
Others				
Year				
2014	0.738[− 0.623, 2.157]*			
2015	0.006[− 0.423, 0.746]	0.453[− 0.684, 0.920]*	0.386[− 0.411, 1.980]+	− 0.179[− 0.319, 0.066]*
2016	− 0.043[− 0.191, 0.463]	0.411[− 0.006, 0.661]**	− 0.163[− 0.328, 0.404]	− 0.060[− 0.250, 0.165]
2017	− 0.006[− 0.234, 0.244]	0.289[− 0.001, 0.488]**	− 0.086[− 0.187, 0.291]	0.108[− 0.052, 0.195]+
2018	0.084[− 0.208, 0.270]	0.334[− 0.015, 0.391]***	− 0.057[− 0.145, 0.260]	0.086[− 0.021, 0.203]+
2019	–	–	–	–
QIC	36, 320	18, 536	12, 994	34, 299
QICu	9485	4853	3205	9215
N	9486	4865	3223	9227

Coefficients were estimated from the whole sample, while BCIs were obtained from 500 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Table 6 Results of discrete time recurrent event model with 2000 bootstraps samples

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th & + vs. 4th Beta[2.5th, 97.5th] BCI
Intercept	− 4.402[− 4.274, − 3.651]***#	− 4.451[− 4.404, − 3.618]***#	− 4.486[− 4.535, − 3.738]***#	− 4.502[− 4.703, − 4.224]***#
Network				
Agent	− 0.375[− 0.286, 0.145]*	− 0.609[− 0.774, − 0.123]*#	− 0.196[− 0.751, − 0.009]#	0.212[− 0.036, 0.472]
Puppet	0.068[− 0.154, 0.100]	0.146[− 0.209, 0.032]	− 0.164[− 0.068, 0.262]+	− 0.022[− 0.243, − 0.057]#
Non-broker	− 0.974[− 2.011, − 1.010]*#	− 1.018[− 1.098, − 0.367]**#	− 1.801[− 2.040, − 0.867]***#	− 0.774[− 1.031, − 0.617]***#
Shared phone	0.491[0.359, 0.568]***#	0.618[0.447, 0.648]***#	0.708[0.555, 0.726]***#	0.527[0.468, 0.567]***#
Coherence	0.876[0.750, 1.644]+#	0.842[0.322, 0.981]**#	1.632[0.619, 1.664]***#	0.630[0.501, 0.911]***#
Message				
Msg senti	− 0.041[− 0.038, 0.055]	− 0.009[− 0.017, 0.102]	− 0.105[− 0.105, − 0.007]**#	0.012[− 0.001, 0.059]
# of replies	− 0.011[− 0.038, 0.020]	− 0.026[− 0.018, 0.035]+	0.000[− 0.026, 0.026]	− 0.013[− 0.026, 0.007]
# of self replies	0.229[− 0.048, 0.310]*	0.175[− 0.109, 0.209]	0.113[− 0.371, 0.098]	0.192[− 0.083, 0.150]+
Reply senti 1std	0.007[− 0.016, 0.118]	0.148[0.029, 0.177]**#	− 0.143[− 0.141, − 0.019]**#	− 0.051[− 0.088, − 0.017]+#
Time stamp				
Day 7	1.216[0.509, 0.934]***#	0.967[0.438, 0.862]***#	0.702[0.157, 0.742]**#	0.798[0.703, 1.015]***#
14	0.705[− 0.051, 0.547]***	0.166[− 0.713, 0.139]	0.607[0.117, 0.805]+#	0.212[0.048, 0.528]#
21	0.481[− 0.501, 0.312]+	− 0.204[− 0.776, 0.228]	0.341[− 0.281, 0.709]	0.255[− 0.022, 0.524]
Controls				
Device				
Cell	− 0.039[− 0.175, − 0.050]#	− 0.129[− 0.131, − 0.004]*#	− 0.033[− 0.077, 0.071]	0.033[− 0.028, 0.070]
Others				
Location				
Hotel	− 0.146[− 0.200, − 0.054]+#	− 0.002[− 0.143, − 0.001]#	− 0.028[− 0.027, 0.138]	− 0.037[− 0.063, 0.023]

Table 6 (continued)

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th & + vs. 4th Beta[2.5th, 97.5th] BCI
Others				
Year				
2014	0.738[− 0.173, 1.830]*			
2015	0.006[− 0.161, 0.381]	0.453[− 0.165, 0.466]*	0.386[− 0.078, 0.827]+	− 0.179[− 0.217, − 0.039]*#
2016	− 0.043[− 0.022, 0.273]	0.411[0.212, 0.539]**#	− 0.163[− 0.159, 0.183]	− 0.060[− 0.129, 0.051]
2017	− 0.006[− 0.090, 0.127]	0.289[0.158, 0.384]**#	− 0.086[− 0.066, 0.155]	0.108[− 0.003, 0.129]+
2018	0.084[− 0.054, 0.163]	0.334[0.133, 0.308]***#	− 0.057[− 0.038, 0.152]	0.086[0.039, 0.146]+#
2019	–	–	–	–
QIC	36, 320	18, 536	12, 994	34, 299
QICu	9485	4853	3205	9215
N	9486	4865	3223	9227

Coefficients were estimated from the whole sample, while BCIs were obtained from 2000 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Table 7 Results of discrete time recurrent event model with 3000 bootstraps samples

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th & + vs. 4th Beta[2.5th, 97.5th] BCI
Intercept	− 4.402[− 4.180, − 3.684]***#	− 4.451[− 4.302, − 3.728]***#	− 4.486[− 4.446, − 3.806]***#	− 4.502[− 4.669, − 4.259]***#
Network				
Agent	− 0.375[− 0.244, 0.102]*	− 0.609[− 0.706, − 0.181]*#	− 0.196[− 0.691, − 0.067]#	0.212[0.000, 0.423]
Puppet	0.068[− 0.133, 0.082]	0.146[− 0.189, 0.006]	− 0.164[− 0.023, 0.239]+	− 0.022[− 0.229, − 0.073]#
Non-broker	− 0.974[− 1.915, − 1.124]*#	− 1.018[− 1.027, − 0.468]**#	− 1.801[− 1.941, − 1.014]***#	− 0.774[− 1.002, − 0.654]***#
Shared phone	0.491[0.365, 0.533]***#	0.618[0.460, 0.622]***#	0.708[0.559, 0.702]***#	0.527[0.477, 0.559]***#
Coherence	0.876[0.836, 1.546]+#	0.842[0.395, 0.915]**#	1.632[0.751, 1.580]***#	0.630[0.543, 0.863]***#
Message				
Msg senti	− 0.041[− 0.026, 0.050]	− 0.009[− 0.005, 0.089]	− 0.105[− 0.093, − 0.016]**#	0.012[0.003, 0.055]#
# of replies	− 0.011[− 0.034, 0.011]	− 0.026[− 0.014, 0.028]+	0.000[− 0.020, 0.022]	− 0.013[− 0.024, 0.002]
# of self replies	0.229[− 0.019, 0.268]*	0.175[− 0.090, 0.168]	0.113[− 0.311, 0.066]	0.192[− 0.063, 0.123]+
Reply senti 1std	0.007[− 0.008, 0.108]	0.148[0.047, 0.164]**#	− 0.143[− 0.133, − 0.030]**#	− 0.051[− 0.082, − 0.026]+#
Time stamp				
Day 7	1.216[0.551, 0.899]***#	0.967[0.477, 0.831]***#	0.702[0.231, 0.715]**#	0.798[0.732, 0.987]***#
14	0.705[0.016, 0.496]***#	0.166[− 0.612, 0.087]	0.607[0.208, 0.761]+#	0.212[0.116, 0.492]#
21	0.481[− 0.408, 0.254]+	− 0.204[− 0.673, 0.144]	0.341[− 0.148, 0.638]	0.255[0.042, 0.497]#
Controls				
Device				
Cell	− 0.039[− 0.164, − 0.060]#	− 0.129[− 0.118, − 0.015]*#	− 0.033[− 0.062, 0.055]	0.033[− 0.018, 0.059]
Others				
Location				
Hotel	− 0.146[− 0.187, − 0.068]+#	− 0.002[− 0.130, − 0.017]#	− 0.028[− 0.014, 0.120]	− 0.037[− 0.055, 0.012]

Table 7 (continued)

Parameters	2nd vs. 1st Beta[2.5th, 97.5th]BCI	3rd vs. 2nd Beta[2.5th, 97.5th] BCI	4th vs. 3rd Beta[2.5th, 97.5th] BCI	5th & + vs. 4th Beta[2.5th, 97.5th] BCI
Others				
Year				
2014	0.738[− 0.023, 1.612]*			
2015	0.006[− 0.104, 0.328]	0.453[− 0.103, 0.373]*	0.386[− 0.005, 0.710]+	− 0.179[− 0.200, − 0.053]*#
2016	− 0.043[0.008, 0.253]#	0.411[0.254, 0.499]**#	− 0.163[− 0.134, 0.149]	− 0.060[− 0.107, 0.037]
2017	− 0.006[− 0.077, 0.109]	0.289[0.189, 0.364]**#	− 0.086[− 0.049, 0.135]	0.108[0.012, 0.116]+#
2018	0.084[− 0.035, 0.145]	0.334[0.155, 0.300]***#	− 0.057[− 0.022, 0.138]	0.086[0.053, 0.137]+#
2019	–	–	–	–
QIC	36, 320	18, 536	12, 994	34, 299
QICu	9485	4853	3205	9215
N	9486	4865	3223	9227

Coefficients were estimated from the whole sample, while BCIs were obtained from 3000 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Table 8 Results of discrete time recurrent event model with 5000 bootstraps samples

Parameters	2nd vs 1st Beta[2.5th, 97.5th]BCI	3rd vs 2nd Beta[2.5th, 97.5th] BCI	4th vs 3rd Beta[2.5th, 97.5th] BCI	5th & + vs 4th Beta[2.5th, 97.5th] BCI
Intercept	− 4.402[− 4.111, − 3.750]***#	− 4.451[− 4.252, − 3.785]***#	− 4.486[− 4.359, − 3.887]***#	− 4.502[− 4.631, − 4.322]***#
Network				
Agent	− 0.375[− 0.203, 0.059]*	− 0.609[− 0.642, − 0.228]*#	− 0.196[− 0.598, − 0.144]#	0.212[0.066, 0.390]#
Puppet	0.068[− 0.112, 0.057]	0.146[− 0.161, − 0.015]#	− 0.164[0.003, 0.206]+#	− 0.022[− 0.214, − 0.088]#
Non-broker	− 0.974[− 1.824, − 1.223]*#	− 1.018[− 0.986, − 0.511]**#	− 1.801[− 1.857, − 1.157]***#	− 0.774[− 0.965, − 0.697]***#
Shared phone	0.491[0.376, 0.496]***#	0.618[0.474, 0.598]***#	0.708[0.573, 0.678]***#	0.527[0.483, 0.546]***#
Coherence	0.876[0.936, 1.479]+#	0.842[0.438, 0.861]**#	1.632[0.836, 1.516]***#	0.630[0.587, 0.837]***#
Message				
Msg senti	− 0.041[− 0.019, 0.039]	− 0.009[0.002, 0.080]#	− 0.105[− 0.085, − 0.026]**#	0.012[0.010, 0.050]#
# of replies	− 0.011[− 0.030, 0.007]	− 0.026[− 0.010, 0.022]+	0.000[− 0.016, 0.014]	− 0.013[− 0.021, − 0.002]#
# of self replies	0.229[0.009, 0.249]*#	0.175[− 0.056, 0.135]	0.113[− 0.245, 0.021]	0.192[− 0.042, 0.104]+
Reply senti 1st d	0.007[0.014, 0.095]#	0.148[0.061, 0.149]**#	− 0.143[− 0.118, − 0.043]**#	− 0.051[− 0.075, − 0.031]+#
Time stamp				
Day 7	1.216[0.597, 0.857]***#	0.967[0.524, 0.808]***#	0.702[0.288, 0.656]**#	0.798[0.754, 0.950]***#
14	0.705[0.085, 0.455]***#	0.166[− 0.515, 0.012]	0.607[0.265, 0.718]+#	0.212[0.151, 0.450]#
21	0.481[− 0.318, 0.183]+	− 0.204[− 0.536, 0.072]	0.341[− 0.032, 0.546]	0.255[0.101, 0.441]#
Controls				
Device				
Cell	− 0.039[− 0.153, − 0.073]#	− 0.129[− 0.109, − 0.031]*#	− 0.033[− 0.047, 0.047]	0.033[− 0.007, 0.052]
Others				
Location				
Hotel	− 0.146[− 0.173, − 0.085]+#	− 0.002[− 0.116, − 0.026]#	− 0.028[0.002, 0.103]#	− 0.037[− 0.046, 0.007]

Table 8 (continued)

Parameters	2nd vs 1st Beta[2.5th, 97.5th]BCI	3rd vs 2nd Beta[2.5th, 97.5th] BCI	4th vs 3rd Beta[2.5th, 97.5th] BCI	5th & + vs 4th Beta[2.5th, 97.5th] BCI
Others				
Year				
2014	0.738[0.128, 1.263]*#			
2015	0.006[− 0.049, 0.270]	0.453[− 0.050, 0.344]*	0.386[0.056, 0.621]+#	− 0.179[− 0.187, − 0.071]*#
2016	− 0.043[0.043, 0.229]#	0.411[0.273, 0.479]**#	− 0.163[− 0.098, 0.109]	− 0.060[− 0.090, 0.021]
2017	− 0.006[− 0.051, 0.086]	0.289[0.204, 0.345]**#	− 0.086[− 0.028, 0.109]	0.108[0.022, 0.102]+#
2018	0.084[− 0.004, 0.128]	0.334[0.164, 0.280]***#	− 0.057[− 0.001, 0.122]	0.086[0.060, 0.127]+#
2019	–	–	–	–
QIC	36, 320	18, 536	12, 994	34, 299
QICu	9485	4853	3205	9215
N	9486	4865	3223	9227

Coefficients were estimated from the whole sample, while BCIs were obtained from 5000 Bootstrapping replicates with size of 1000

#Indicates the 95% BCIs do not include zero; and Model-based p values were indicated by: $+p<.1$, $*p<.05$, $**p<.01$, $***p<.001$

Appendix

See Tables 3, 4, 5, 6, 7 and 8.

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Data availability No data associated in the manuscript.

Declarations

Conflict of interest The authors have no conflicts of interest to declare. All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication.

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